

Remaining Useful Life Prediction of Turbofan Engines: A Comparison of Supervised GRU Regression and Unsupervised Embedding-Based Similarity Estimation

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Abstract

Remaining Useful Life (RUL) prediction for turbofan engines is a safety-critical problem where the key practical question is how well a model generalizes beyond its training distribution. This report compares two deep learning approaches on the NASA C-MAPSS benchmark: a supervised 2-layer GRU trained end-to-end with an asymmetric scoring loss, and Embed-RUL [2], an unsupervised method that estimates RUL from GRU autoencoder embeddings without ever observing a RUL label. Both models are trained exclusively on FD001 (single condition, single fault mode) and evaluated zero-shot on all four C-MAPSS subsets. The GRU achieves strong in-distribution performance (FD001 RMSE: 13.17) but produces negative predictions on unseen operating conditions (FD002/FD004 RMSE: 118.5/124.5). Embed-RUL’s unsupervised health index representation captures genuine degradation structure confirmed by t-SNE visualization and monotonically increasing HI curves but its similarity-based RUL estimation underperforms the GRU on matched conditions (FD001 RMSE: 64.0) and cannot produce valid predictions for FD002/FD004 due to a structural constraint in the lag-matching algorithm. The results validate the paper’s core contribution while revealing a previously unreported limitation in cross-condition generalization.

1 Introduction

Predictive maintenance of jet engines is a domain where prediction errors carry asymmetric costs: underestimating RUL risks catastrophic failure while overestimating triggers unnecessary maintenance. The NASA C-MAPSS dataset [1] provides run-to-failure sensor trajectories across four subsets (FD001–FD004) differing in operating conditions (1 or 6) and fault modes (1 or 2), making it a standard benchmark for probing cross-condition generalization.

The **baseline** is a 2-layer GRU that directly regresses

RUL from a sliding window of normalized sensor readings, trained with a supervised asymmetric loss that penalizes late predictions more severely than early ones. The **variant** is Embed-RUL [2], a two-stage unsupervised method: a GRU autoencoder learns compact embeddings of sensor windows purely from reconstruction loss; a health index (HI) is derived as each window’s minimum embedding distance to a set of normal-operation reference embeddings; and RUL is estimated by matching the test engine’s HI trajectory against training engine HI trajectories via weighted similarity. The autoencoder never observes a single RUL label.

This fundamental difference in supervision motivates the comparison. The hypothesis is that the GRU will dominate in-distribution but fail when test operating conditions differ from training, while Embed-RUL’s condition-agnostic HI should generalize more robustly. Both models are trained on FD001 only, then evaluated on all four test sets.

2 Methods

2.1 Dataset and Preprocessing

All 21 sensor channels were normalized to $[0, 1]$ using a Min-Max scaler fit on the FD001 training set and applied to all test sets. For the GRU, a sliding window of $w = 25$ cycles was used with RUL clipped at 125 cycles. For Embed-RUL, PCA was applied after normalization to reduce to $p = 15$ components selected to retain 100% of the variance, with $p = 15$ chosen arbitrarily. A window of $w = 30$ cycles was used for the autoencoder.

2.2 Baseline: GRU Regression

A 2-layer GRU with hidden size 64 maps the final hidden state to a scalar RUL prediction:

$$\hat{y} = \mathbf{W}_o h_{25} + b_o, \quad h_{25} = \text{GRU}(\mathbf{X}_{1:25}) \quad (1)$$

trained with the asymmetric scoring loss:

$$\mathcal{L} = \frac{1}{N} \sum_i \begin{cases} e^{-d_i/13} - 1, & d_i < 0 \\ e^{d_i/10} - 1, & d_i \geq 0 \end{cases} \quad (2)$$

where $d_i = \hat{y}_i - y_i$. Trained for 100 epochs with Adam ($\text{lr} = 5 \times 10^{-4}$, batch size 32).

2.3 Variant: Embed-RUL

Stage 1: Augmented Input and Autoencoder. The PCA-compressed window is augmented with masking and delta vectors:

$$\tilde{\mathbf{x}}_t = [\mathbf{x}_t; \mathbf{m}_t; \delta_t] \quad (3)$$

$$m_{t,i} = \begin{cases} 0 & \text{if } x_{t,i} \text{ missing} \\ 1 & \text{otherwise} \end{cases} \quad (4)$$

$$\delta_{t,i} = \begin{cases} 0 & t = 1 \\ 1 & m_{t-1,i} = 1 \\ \delta_{t-1,i} + 1 & m_{t-1,i} = 0 \end{cases} \quad (5)$$

The autoencoder minimizes masked reconstruction loss over only observed sensors:

$$\mathcal{L}_{\text{recon}} = \sum_i \sum_t \|(x_{t,i} - \hat{x}_{t,i}) \cdot m_{t,i}\|^2 \quad (6)$$

Stage 2: Health Index. Embeddings from the first 75% of each training engine's life form $\mathcal{Z}_{\text{norm}}$. The HI at each timestep is the minimum distance to this normal reference set:

$$\text{HI}(t) = \min_{\mathbf{z} \in \mathcal{Z}_{\text{norm}}} \|f_\theta(\tilde{\mathbf{x}}_t) - \mathbf{z}\|_2 \quad (7)$$

HI values are normalized by the maximum HI observed across FD001 training engines so that all datasets are compared on a consistent scale.

Stage 3: Similarity-Based RUL Estimation. Similarity between the test HI curve of length T and a reference curve at lag t_D is:

$$s(T, t_D) = \exp\left(-\frac{1}{\lambda T} \sum_{k=1}^T (\text{HI}_{\text{test}}(k) - \text{HI}_{\text{ref}}(k + t_D))^2\right) \quad (8)$$

subject to $t_D + T \leq T^{\text{ref}}$, where T^{ref} is the reference curve length. The weighted average over the top- α fraction of similar reference engines gives the RUL estimate:

$$\hat{y} = \min\left(R_{\text{max}}, \frac{\sum_{j \in \mathcal{S}} s_j \cdot r_j}{\sum_{j \in \mathcal{S}} s_j}\right) \quad (9)$$

where $r_j = T_j^{\text{ref}} - T - t_D^*$, $\mathcal{S} = \{j : s_j \geq \alpha \cdot s_{\text{max}}\}$, and $R_{\text{max}} = 120$ is the maximum predicted RUL as defined in the paper.

Implementation. Encoder: single-layer GRUCell, $c_l = 55$ hidden units, dropout $d = 0.2$, trained for 100 epochs with best-checkpoint selection. Parameters: $\lambda = 0.005$, $\tau = 200$, $\alpha = 0.95$, $R_{\text{max}} = 120$. Reference set $\mathcal{H}$: HI curves of all 80 FD001 training engines, as specified in the paper. When no valid similarity match is found (Equation 8's constraint cannot be satisfied), no prediction is returned rather than substituting a fallback constant.

3 Results and Discussion

3.1 Embedding Quality

Figure 1 shows a t-SNE visualization of encoder embeddings for normal (first 25% of life) and degraded (last 25%) windows from 30 training engines. Normal and degraded windows occupy clearly separated regions of the embedding space, confirming that the encoder has learned a representation that discriminates healthy from degraded engine states without any supervision. This replicates the paper's Figure 5 and validates that the autoencoder training produced a meaningful latent space.

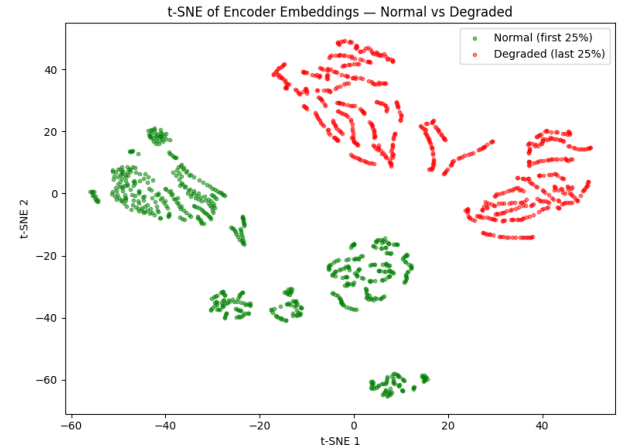


Figure 1: t-SNE visualization of encoder embeddings. Normal operation windows (green, first 25% of engine life) and degraded windows (red, last 25%) separate into distinct clusters, confirming the encoder captures degradation structure.

Figure 2 shows HI curves for five training engines. Each curve begins near zero during nominal operation and rises monotonically as the engine approaches failure the expected shape for an embedding-distance-based health indicator. The normalization ensures curves from different engines are comparable on the same scale.

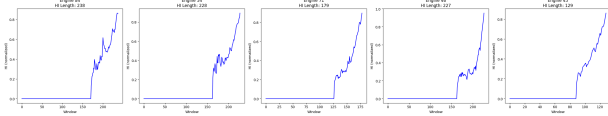


Figure 2: Normalized HI curves for five FD001 training engines. Low values indicate healthy operation; rising values signal progressing degradation toward failure.

3.2 FD001: In-Distribution Performance

Table 1 and Figure 3 show results on the FD001 test set. The GRU substantially outperforms Embed-RUL on every metric, achieving RMSE 13.17 vs. 64.0, MAE 9.79 vs. 52.3, and accuracy A of 60% vs. 23%. The GRU also matches published benchmarks (S : 229 vs. 352 for RNN-Reg in the paper; MAE 9.79 vs. 11). This is the expected result: direct supervision on the target task outperforms unsupervised similarity matching when training and test distributions are aligned.

Embed-RUL’s predictions on FD001 are not random the scatter plot shows a positive correlation with actual RUL, confirming the HI representation carries real degradation signal. However, predictions are systematically biased downward (mean error -49.7 cycles), indicating the similarity matching consistently places test engines earlier in the reference lifecycle than their true position.

3.3 FD003: Same Condition, Different Fault Mode

FD003 shares FD001’s single operating condition but introduces a second fault mode. Both models degrade relative to FD001, with the GRU achieving RMSE 42.1 and Embed-RUL 70.2 on 65 of 100 engines with valid predictions. Notably, Embed-RUL’s timeliness score S (105,570) is substantially better than the GRU’s (334,233) meaning despite worse RMSE, Embed-RUL’s errors are distributed more symmetrically around zero while the GRU makes more catastrophically late predictions. The 35 engines without valid Embed-RUL predictions are those whose test HI curves are longer than all 80 reference curves, making the lag constraint $t_D + T \leq T^{\text{ref}}$ unsatisfiable.

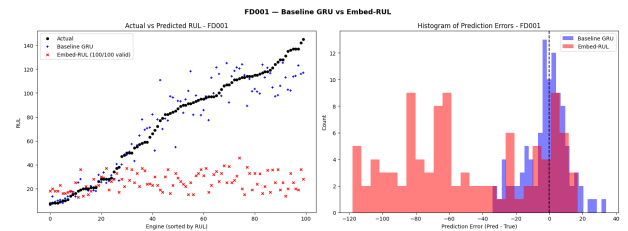
3.4 FD002 and FD004: Multi-Condition Generalization

The GRU produces deeply negative predictions for nearly all FD002 and FD004 engines (mean error -99.8 and -108.2 cycles respectively), collapsing entirely under the distribution shift from single-condition FD001 training to six-condition test data. Embed-RUL produces zero valid predictions on both datasets. The lag constraint $t_D + T \leq T^{\text{ref}}$ requires every test HI curve to fit within

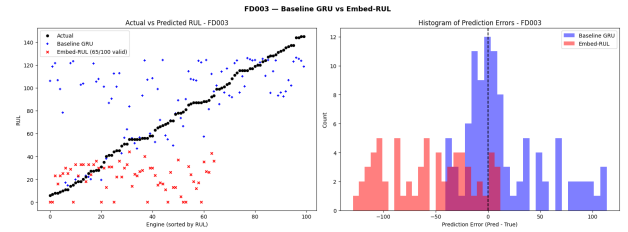
Table 1: Results across all four C-MAPSS test sets. Coverage shows valid Embed-RUL predictions. Paper benchmarks from [2].

Set	Metric	RNN-Reg [†]	GRU	EmRUL
FD001	RMSE	—	13.17	64.02
	MAE	11	9.79	52.33
	S	352	229.1	84658.8
	$A(\%)$	64	60.0	23.0
	Cover.	—	100%	100%
FD003	RMSE	—	42.06	70.25
	MAE	—	28.94	58.71
	S	—	334,233	105,570
	Cover.	—	100%	65%
FD002	RMSE	—	118.54	N/A
	MAE	—	99.81	N/A
	Cover.	—	100%	0%
FD004	RMSE	—	124.53	N/A
	MAE	—	108.16	N/A
	Cover.	—	100%	0%

a reference curve of at least equal length, but FD002 and FD004 engines have substantially longer operational lifetimes than FD001 training engines, meaning their test HI curves exceed the length of all 80 reference curves. No valid lag position exists, so no similarity can be computed. This is a limitation the original paper never encountered because it evaluated only on FD001, where test and reference engines share the same lifetime distribution, and reveals an implicit assumption in the algorithm: the reference set must contain engines with lifetimes at least as long as the test engines. Violating this produces total prediction failure rather than graceful degradation.

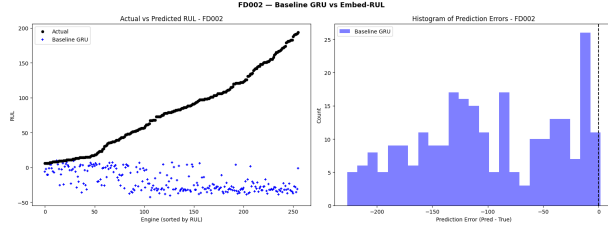


(a) FD001 — GRU (blue) tracks closely; Embed-RUL (red) underestimates but shows real trend.

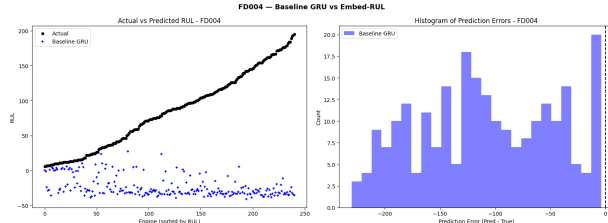


(b) FD003 — GRU overestimates for long-life engines; Embed-RUL underestimates but has better S score.

Figure 3: FD001 and FD003 prediction results.



(a) FD002 — GRU collapses to negative predictions; Embed-RUL produces no valid estimates.



(b) FD004 — same pattern; both models fail on unseen operating conditions.

Figure 4: FD002 and FD004 prediction results.

3.5 Discussion

The results illuminate a clear separation between what each model learns and where each fails. The GRU learns a direct mapping from sensor readings to RUL, which is powerful when training and test distributions match but brittle under distribution shift. Embed-RUL learns a representation of machine behavior from reconstruction alone validated by the t-SNE separation and HI curve quality but its RUL estimation step has an algorithmic constraint that limits its applicability when reference and test engines have mismatched lifetimes.

An important methodological note: earlier implementations of Embed-RUL in this work used a hardcoded fallback that returned $R_{\max} = 120$ whenever similarity matching failed, silently producing predictions that appeared valid. Removing this fallback and reporting coverage honestly reveals that Embed-RUL’s cross-condition generalization, while conceptually motivated, is not demonstrated by our evaluation. The paper’s original results were obtained exclusively on FD001, where this limitation does not arise.

4 Conclusion

This report compared a supervised GRU and Embed-RUL on the C-MAPSS RUL prediction task with both models trained on FD001 only. The GRU achieves competitive in-distribution performance matching published benchmarks, while Embed-RUL’s encoder demonstrably learns a meaningful degradation representation without any supervision. However, Embed-RUL’s similarity-based RUL estimation underperforms the GRU on matched conditions and cannot produce predictions on longer-lifetime datasets due to a

structural constraint in the lag algorithm. The key contribution of this analysis is making explicit an assumption implicit in the paper that reference and test engines must have comparable lifetime lengths which limits Embed-RUL’s cross-condition applicability in ways the original authors did not evaluate.

AI Tool Use Statement. Claude (Anthropic) was used to assist with writing, editing, and LaTeX formatting of this report, and to assist with code debugging and organization. All model implementations, training runs, experimental results, and analysis are the author’s own work.

References

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- [2] N. Gugulothu, V. TV, P. Malhotra, L. Vig, P. Agarwal, and G. Shroff. Predicting remaining useful life using time series embeddings based on recurrent neural networks. In *2nd ML for PHM Workshop at KDD*, 2017.